COURSE END PROJECT

SMART GAS LEAKAGE DETECTION SYSTEM

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VARDHAMAN COLLEGE OF ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

This is to certify that the Course End Project titled **"Smart Gas Leakage Detection System "** is carried out by **Allam Akshaya , M.Nisha Reddy** with Roll Numbers **24881A05D2, 24881A05G7** towards **A8603 - IOT Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** during the Academic year 2025-26.

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Check List

S. No.	Content	Status (✓ / ×)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Working model	✓
9	Conclusion	✓
10	Future Scope	✓
11	References	✓

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1.ABSTRACT

The Smart Gas Leakage Detection System is an Internet of Things (IoT)-based safety solution designed to detect hazardous gases such as LPG, methane, and smoke in real time. Gas leakage poses significant risks, including fire accidents, explosions, and serious health hazards, making early detection crucial for ensuring safety.

This project focuses on developing a simple, reliable, and cost-effective system that continuously monitors gas levels and provides immediate alerts when unsafe conditions are detected. The system utilizes an Arduino Uno microcontroller in conjunction with an MQ-2 gas sensor to measure gas concentration in the surrounding environment. The sensor continuously transmits data to the Arduino, which processes the readings and compares them against a predefined threshold value.

When the detected gas level exceeds the safe limit, the system triggers a buzzer to provide an audible warning and activates an LED for visual indication. This dual-alert mechanism ensures that users are promptly informed about potential gas leaks, allowing them to take necessary preventive measures.

The proposed system is easy to implement and suitable for various applications, including homes, kitchens, laboratories, and small-scale industries. It effectively demonstrates how embedded systems and IoT technologies can be utilized to enhance safety and minimize risks. Furthermore, the system can be extended by integrating wireless communication modules such as GSM or Wi-Fi, enabling remote monitoring and real-time alerts, thereby contributing to smart home and industrial safety solutions.

Overall, this project offers an efficient, practical, and scalable approach to gas leakage detection and prevention.

KEYWORDS: IoT, Arduino Uno, MQ-2 Sensor, Gas Leakage Detection, LPG, Methane, Smoke Detection, Buzzer Alert, LED Indicator, Embedded Systems, Smart Homes, Industrial Safety, Remote Monitoring, Sustainable Development Goals.

2. PROBLEM STATEMENT

Gas leakage is a serious safety concern in residential, commercial, and industrial environments, as it can lead to fire accidents, explosions, and severe health hazards. Hazardous gases such as LPG and methane are widely used but are highly flammable and difficult to detect without proper sensing mechanisms. Even a small undetected leak can quickly escalate into a dangerous situation, especially in enclosed spaces.

Traditional gas detection methods are not reliable enough to ensure safety. Most of them depend on human perception, such as smell, or on simple alarm systems that may not respond immediately. These approaches often result in delayed detection, increasing the risk of accidents and damage. Additionally, they lack continuous monitoring and automation, which are essential for modern safety systems.

Although advanced gas detection systems are available, they are often expensive, complex, and require technical expertise for installation and maintenance. This makes them unsuitable for common households, small laboratories, and small-scale industries. As a result, many environments still lack effective and affordable safety solutions for gas leakage detection.

With the advancement of embedded systems and Internet of Things (IoT) technologies, there is a growing need for intelligent systems that can continuously monitor environmental conditions and provide real-time alerts. Such systems should be capable of detecting hazardous gases accurately, responding instantly to unsafe conditions, and minimizing human intervention.

Therefore, the problem addressed in this project is to design and develop a **smart, cost-effective, and efficient gas leakage detection system** that continuously monitors gas levels, detects hazardous conditions in real time, and provides immediate alerts through audible and visual indicators. The system should be reliable, easy to implement, and suitable for practical applications, while also allowing future enhancements such as remote monitoring and smart home integration.

3. OBJECTIVES

The primary objective of this project is to design and develop a smart gas leakage detection system that enhances safety by continuously monitoring the presence of hazardous gases in the surrounding environment. The system aims to make use of embedded systems and IoT concepts to provide real-time detection and immediate alerts, thereby reducing the risk of fire accidents, explosions, and health hazards caused by gas leaks. It focuses on building a solution that is simple, reliable, cost-effective, and easy to deploy in practical environments such as homes, kitchens, laboratories, and small-scale industries. Additionally, the project seeks to demonstrate how modern technology can be effectively used to address real-world safety problems and improve overall awareness and responsiveness.

- To design and develop a smart gas leakage detection system using embedded systems and IoT technologies.
- To continuously monitor gas concentration in the environment using the MQ-2 sensor for real-time detection.
- To accurately detect hazardous gases such as LPG, methane, and smoke under different environmental conditions.
- To provide immediate alerts through a buzzer (audible warning) and LED (visual indication) when gas levels exceed predefined safety thresholds.
- To ensure user safety by enabling early detection of gas leaks and allowing timely preventive actions.
- To develop a system that is cost-effective, reliable, energy-efficient, and easy to implement for real-time applications.
- To gain practical knowledge in sensor interfacing, microcontroller programming, and embedded system design.
- To explore the possibility of integrating advanced features such as IoT-based remote monitoring and smart home automation in future enhancements

4. Algorithm

The following steps describe the working of the Smart Gas Leakage Detection System:

Start the system.

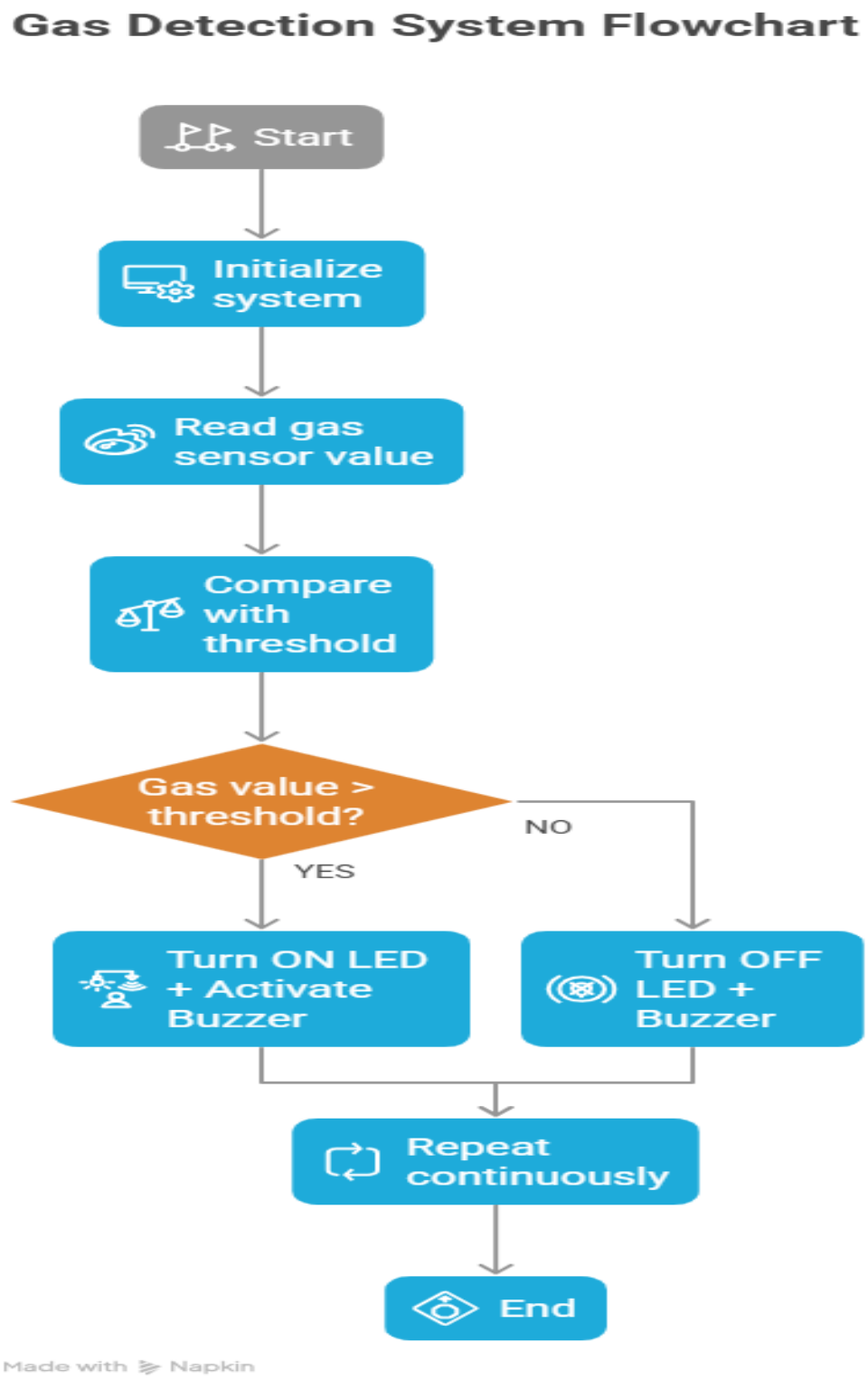
1. Initialize the input and output pins:
2. Set the gas sensor pin (A0) as input.
3. Set the buzzer and LED pins as output.
4. Initialize serial communication for monitoring gas values.
5. Allow the gas sensor to stabilize (warm-up period).
6. Continuously read the gas concentration value from the MQ-2 sensor.
7. Display the gas value on the serial monitor.
8. Compare the sensed gas value with the predefined threshold level.
9. If the gas value exceeds the threshold:
10. Turn ON the LED (visual alert).
11. Activate the buzzer to generate an alarm sound.
12. Repeat the buzzer alert for a few cycles.
13. Else (if gas value is below threshold):
14. Turn OFF the LED.
15. Turn OFF the buzzer.
16. Repeat the process continuously to ensure real-time monitoring.
17. Stop.

5. PROPOSED SYSTEM

The proposed system is a smart gas leakage detection system designed to continuously monitor the presence of hazardous gases such as LPG, methane, and smoke in the environment. The system uses an MQ-2 gas sensor to detect gas concentration and an Arduino Uno microcontroller to process the sensor data. When the gas concentration exceeds a predefined threshold level, the system activates alert mechanisms such as a buzzer and an LED to notify the user immediately.

The system is designed to be simple, cost-effective, and reliable, making it suitable for practical applications in homes, kitchens, laboratories, and small industries. It operates in real time and does not require human intervention, thereby reducing the chances of accidents caused by delayed detection. The modular design also allows future enhancements such as IoT-based remote monitoring and automated safety mechanisms.

5.1 FLOW CHART



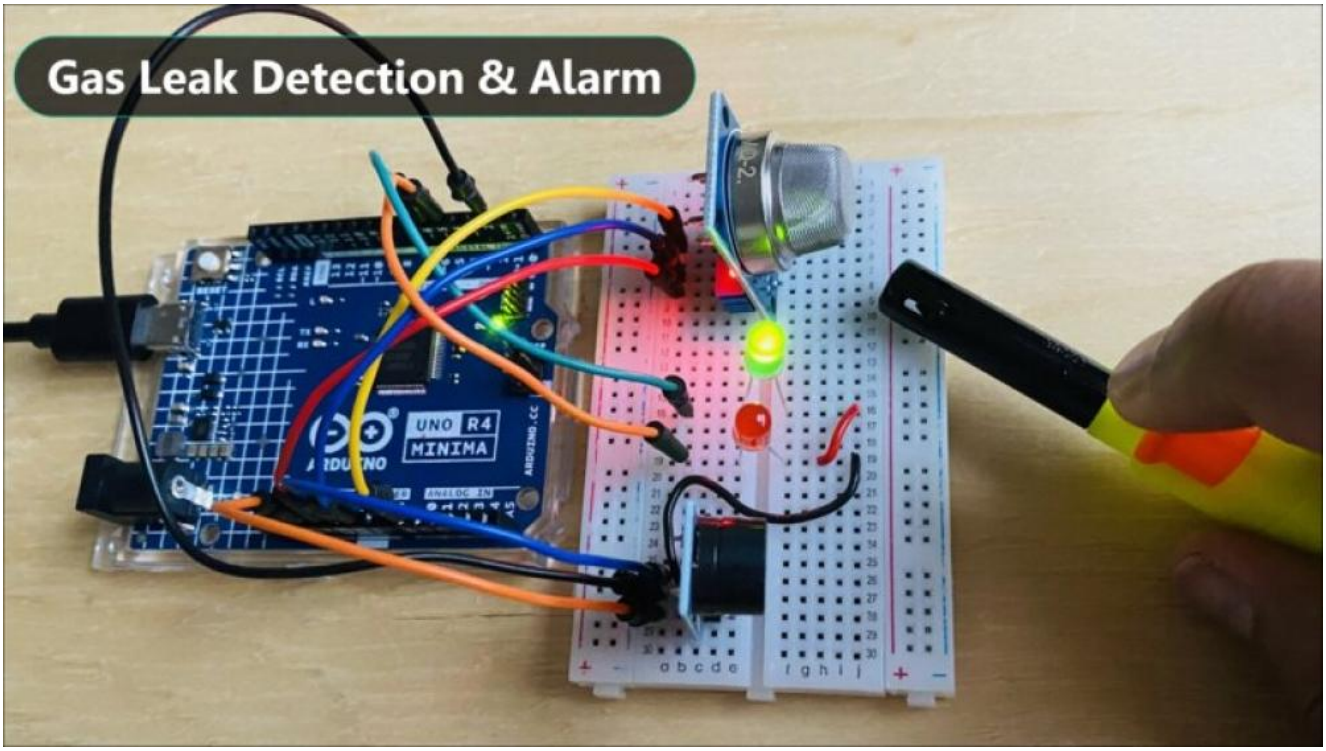
5.2 WORKING MODEL

The working model of the system is based on real-time gas sensing and alert generation. The MQ-2 gas sensor continuously monitors the surrounding environment and detects the presence of gases such as LPG, methane, and smoke. It converts the gas concentration into an analog signal, which is sent to the Arduino microcontroller.

The Arduino reads the sensor value through its analog input pin and compares it with a predefined threshold value. If the gas concentration exceeds the safe limit, the system identifies it as a gas leakage condition. In response, the Arduino activates a buzzer to produce an audible alert and turns on an LED to provide a visual indication.

If the gas level is within the safe range, the system remains inactive, with both the buzzer and LED turned off. This process is repeated continuously, ensuring real-time monitoring and quick response to hazardous conditions.

The system is simple to implement and demonstrates the effective use of embedded systems for safety applications. It can be further enhanced by integrating communication modules such as Wi-Fi or GSM for remote alerts and smart monitoring.



6.SOURCE CODE

```
int gasSensor = A0;
int buzzer = 8;
int bulb = 9;
int threshold = 600;
void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(bulb, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int gasValue = analogRead(gasSensor);
  Serial.println(gasValue);

  if (gasValue > threshold) {
    digitalWrite(bulb, HIGH);

    for (int i = 0; i < 3; i++) {
      tone(buzzer, 2000);
      delay(150);
      noTone(buzzer);
      delay(150);
    }

  } else {
    digitalWrite(bulb, LOW);
    noTone(buzzer);
    delay(100);
  }}
```

7.INPUT

The input to the Smart Gas Leakage Detection System is obtained from the MQ-2 gas sensor, which is used to detect the presence and concentration of hazardous gases such as LPG, methane, and smoke in the surrounding environment. The sensor senses gas levels and converts them into an analog electrical signal proportional to the gas concentration.

This analog signal is then provided to the Arduino Uno microcontroller through its analog input pin (A0). The microcontroller continuously reads this input using its built-in Analog-to-Digital Converter (ADC), which converts the analog signal into a digital value ranging from 0 to 1023.
The input data (gas concentration value) is used by the system to determine whether the environment is safe or hazardous by comparing it with a predefined threshold value.

Fig. 2: MQ-2 Gas Sensor

- The MQ-2 sensor detects gases such as LPG, methane, and smoke.
- It sends analog/digital signals to the Arduino.

8.CONCLUSION

The Smart Gas Leakage Detection System developed in this project provides an effective, reliable, and real-time solution for detecting hazardous gases such as LPG, methane, and smoke. By integrating the MQ-2 gas sensor with the Arduino Uno microcontroller, the system continuously monitors gas concentration levels in the environment and compares them with predefined safety thresholds. Whenever the gas level exceeds the safe limit, the system immediately activates alert mechanisms such as a buzzer and an LED, ensuring quick user awareness and enabling timely preventive actions.

This project successfully demonstrates the practical application of embedded systems and IoT concepts in addressing real-world safety challenges. The system is designed to be simple, cost-effective, and easy to implement, making it accessible for a wide range of users including households, laboratories, and small-scale industries. Its ability to operate continuously without human intervention significantly reduces the risk of delayed detection and enhances overall safety.

In addition to its core functionality, the project emphasizes the importance of automation and real-time monitoring in modern safety systems. The use of sensor-based detection ensures higher accuracy and faster response compared to traditional methods. The system also serves as an educational model for understanding sensor interfacing, microcontroller programming, and system design, thereby contributing to practical learning in the field of IoT and embedded systems.

Moreover, the proposed system has significant potential for further improvements and scalability. It can be enhanced by integrating wireless communication technologies such as Wi-Fi or GSM modules to enable remote monitoring and instant notifications through mobile devices. Additional features like automatic gas shut-off mechanisms, cloud-based data logging, and AI-based anomaly detection can further improve the system's efficiency, reliability, and usability in advanced applications.

In conclusion, the Smart Gas Leakage Detection System offers a practical, efficient, and scalable solution for enhancing safety and preventing accidents caused by gas leaks. It not only addresses a critical real-world problem but also lays the foundation for future innovations in smart safety systems, contributing towards safer living environments and sustainable technological development.

9.FUTURE SCOPE

The Smart Gas Leakage Detection System can be further enhanced by integrating advanced technologies to improve its functionality, reliability, and usability. One of the major improvements is the addition of IoT-based communication modules such as Wi-Fi (ESP8266/ESP32) or GSM, which can send real-time alerts to users through mobile applications or SMS. This enables remote monitoring and ensures that users are notified even when they are not physically present near the system.
The system can also be upgraded by incorporating an automatic gas shut-off mechanism, where a solenoid valve can be used to stop the gas supply immediately when leakage is detected. This feature can significantly reduce the chances of fire accidents and explosions by preventing further gas accumulation.

Another enhancement is the integration of display modules such as LCD or OLED screens to show real-time gas concentration levels, making the system more informative and user-friendly. Additionally, the use of multiple gas sensors can improve accuracy and allow detection of a wider range of hazardous gases in different environments.

Further advancements can include cloud-based data storage and analytics, where gas concentration data is recorded and analyzed over time. This can help in identifying patterns, predicting potential risks, and improving system efficiency. Integration with AI or machine learning algorithms can also help in reducing false alarms and improving detection accuracy.

Finally, the system can be expanded into a complete smart home or industrial safety system by integrating it with other sensors and automation technologies. This would enable features such as automatic ventilation, alarm systems, and centralized monitoring, making the solution more robust, scalable, and suitable for real-world deployment.

10.REFERENCES

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Online Resources

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2. TutorialsPoint – Gas Sensor Interfacing with Arduino
3. GeeksforGeeks – IoT Gas Monitoring System
4. Random Nerd Tutorials – MQ-2 Gas Sensor with Arduino
5. Espressif Systems – ESP32 IoT Tutorials